


# Tribhuvan University

## Faculty of Humanities and Social Science

**A Project Report**

**On**

**SIGN LANGUAGE TRANSLATOR**

**Submitted To**

**Department of Computer Application**

**Xavier International College**

**Kathmandu**

***In partial fulfillment of the requirements for the Bachelors in Computer Application***

**Submitted By**

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**Faculty of Humanities and Social Sciences**

**National College of Computer Studies**

**Supervisor's Recommendation**

We hereby recommend that this project prepared under my supervision Milan Bohara and Shraddha Pyakurel entitled "**SIGN LANGUAGE TRANSLATOR** " in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

.....

**SIGNATURE**

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# LETTER OF APPROVAL

This is to certify that this project prepared by Milan Bohara and Shraddha Pyakurel entitled "**SIGN LANGUAGE TRANSLATOR** " in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

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| | |

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# ABSTRACT

According to the World Health Organization (WHO), 466 million people across the world have disabling hearing loss (over 5% of the world's population), of whom 34 million are children. There are only about 250 certified sign language interpreters in India for a deaf population of around 7 million. With these significant statistics, the need for developing a tool for smooth flow of communication between abled and people with speech/hearing impairment is very high. Our application promises to secure a two-way conversation, as it deploys machine learning and deep learning models to convert sign language to text. The opposite receiver can get a text as his response, which will then be visible to the disabled person in the form of text. The client can use the tutorials and learn the basic functioning of the application and ASL. This system eliminates the need for an interpreter, and the traditional methods of pen and paper can also be discarded. This application ensures the automation of communication and thereby provides a solution to the hurdles faced by hearing/speech-impaired people.

***Keywords****: Sign-Language-Translator, Feature-extraction, KNN-Classification, etc.*

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# LIST OF ABBREVIATION

| | |
|---|---|
| **API** | Application Programming Interface |
| **DFD** | Data Flow Diagram |
| **ER** | Entity Relationship |
| **IDE** | Integrate Development Environment |
| **UI** | User Interface |

# CHAPTER 1: INTRODUCTION

## 1.1. Introduction

With the increase of innovations and technology, life has become significantly easier for humans. The sudden surge of growth in tech has left many overjoyed and overwhelmed because of the good fruits it bears. It has paved the way for the poor people to become rich, the sick to become strong, and the disabled to experience the life of an abled. People with speech/hearing impairment have always found it difficult to communicate and mingle but with technology, that barrier has been destroyed. They can now communicate without any difficulty and find themselves in a public setting, communicating confidently.

With the help of technology & internet, we can control and access the machines and things that are connected to the Internet even if the distances are too long. Without human-human and computer-human interaction, we can send and receive information.

## 1.2. Problem Statement

The traditional methods of communicating with the deaf and mute are really not convenient in many aspects. The alternatives that are available to break this barrier have definite flaws. An interpreter is not always available, and this method is not cost-efficient either. The pen-and-paper method is highly unprofessional and also time-consuming. Texting and messaging are fine to a certain extent but still do not tackle the bigger problem at hand. This has created a grave need to develop a solution to destroy the barricade of communication effectively.

## 1.3. Objectives

The primary objectives of the proposed project are:

- To eliminate the need for an interpreter.
- To ease the communication flow for hearing/speech impaired people through our sign-to-text system.
- To create new signs for any text.

## 1.4. Scope and Limitations

Sign Language Translator enables the hearing-impaired user to communicate efficiently in sign language, and the application will translate the sign into text. The user has to train the

model, by recording the sign language gestures and then label the gesture. The user can then use the saved and recorded gestures while speaking to other people through signs.

The major scope of the project includes:

- Explore educational applications to help individuals learn sign language more effectively.
- Use the project for accessibility applications to assist the deaf and hard-of-hearing community in various settings, including educational institutions, workplaces, and public spaces.

There are some limitations to this system. They are listed as follows:

- The user has to provide the dataset by making the sign language gestures and then label them.
- The system may struggle with recognizing gestures in low-light conditions or when the user's hands are partially occluded.
- The system may struggle to recognize complex hand gestures or gestures that involve intricate finger movements, as it relies on predefined landmark points.

## 1.5. Development Methodology

For the development of the project, the waterfall model suits perfectly. The waterfall model is a sequential design process, used in software development processes, in which progress is seen as flowing steadily downwards through the phases of planning, analysis, design, implementation, and testing.

The planning phase is used for understanding why a system should be built and determining how the project team will go about building it. The analysis phase answers the questions of who will use the system, what the system will do, and where and when it will be used. During this analysis phase, the project team investigates any current system, identifies improvement opportunities, and develops a concept for the new system. The design phase decides how the system will operate in terms of the hardware, software, and network infrastructure that will be in place. In the implementation phase, a system is actually created and this phase usually gets the most attention, because for most systems it is the longest and most expensive single part of the development process. The final phase is the testing phase, where the system after implementation needs to be tested to make it error-free.

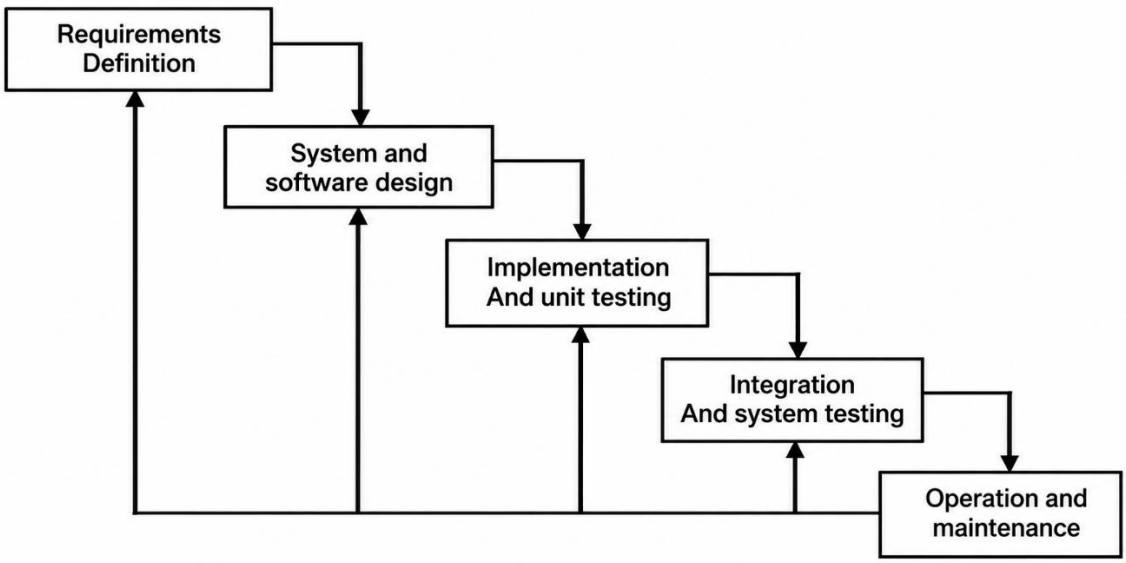



**Figure 1: Waterfall Model**

## 1.6. Report Organization

The report is organized in following chapters:

- **Introduction:** The first chapter defines the basic introduction of the project. It also includes the problem statement, objectives, scope and limitations and development methodology of the system.
- **Background study and Literature review:** The second chapter describes background study and literature review of the existing systems.
- **System analysis and design:** In the third chapter the system analysis and system design are introduced. System analysis includes the requirement analysis, feasibility analysis and various modelling diagrams.
- **Implementation and testing:** The fourth chapter includes tools used for implementation of this project and testing methods and details of the system.
- **Completed and Remaining Works:** The fifth chapter includes all the completed work and remaining works of this project.

# CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

## 2.1. Background Study

Sign language is a system of communication using visual gestures and signs. Sign languages are expressed through manual articulations in combination with non-manual elements. Sign languages are full-fledged natural languages with their own grammar and lexicon. Sign languages are not universal and are usually not mutually intelligible with each other, although there are also similarities among different sign languages.

Using the video input from the webcam Sign Language Translator (SLT) translates the signs into text output. American Sign Language (ASL) is the predominant sign language of Deaf communities in the United States and most of Anglophone Canada. Besides North America, dialects of ASL and ASL-based creoles are used in many countries around the world, including much of West Africa and parts of Southeast Asia.

## 2.2. Literature Review

The first sign-language glove to gain any notoriety came out in 2001. A high-school student from Colorado, Ryan Patterson, fitted a leather golf glove with 10 sensors that monitored finger position, then relayed finger spellings to a computer which rendered them as text on a screen. In 2002, the public affairs office of the National Institute on Deafness and Other Communicative Disorders effused said about Patterson, that the glove doesn't translate anything beyond individual letters, certainly not the full range of signs used in American Sign Language, and works only with the American Manual Alphabet.

https://www.theatlantic.com/technology/archive/2017/11/

Motion Savvy is building a tablet that detects when a person is using ASL and converts it to text or voice. The software also has voice recognition through the tablet's mic, which allows a hearing person to respond with a voice to the person signing

https://techcrunch.com/2014/06/06/motionsavvy-is-a-tablet-app-thatunderstands-sign language/

# CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

## 3.1. System Analysis

The system analysis of our system is as follows:

### 3.1.1. Requirement Analysis

Following requirements were identified to be included in the application.

**i. Functional Requirements**

Functional requirement defines a function of a system or its component, where a function is described as a specification of behavior between outputs and inputs.

| Functional Requirements | Description | Comments |
|---|---|---|
| FR1 | The system must detect the hand signs of the signer. | In order to recognize sign first system must detect hand. |
| FR2 | The system must recognize the hand signs of the signer. | Uses KNN classification to classify and recognize sign. |
| FR3 | The system shall show the output in the form of text or *audio*. | This is the main objective of the project. |

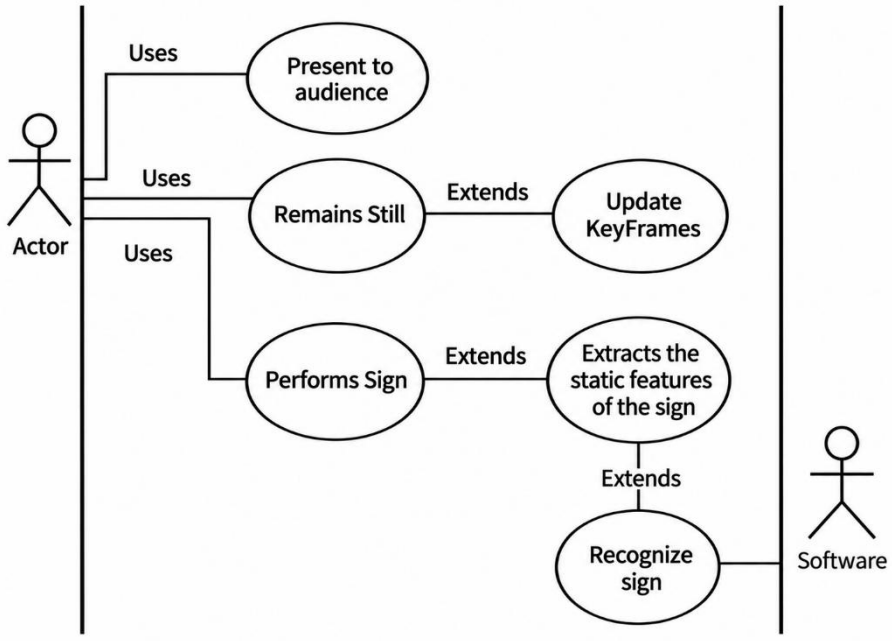



**Figure 1: Use Case Diagram**

**ii. Non-Functional Requirements**

Non-functional Requirements are often called “quality attributes” of a system. Other terms for non-functional requirements are “qualities”, “quality goals”, “quality of service requirements”, “constraints” and “non-behavioral requirements”. Non-functional requirements are:

- **Accuracy:** The information on the user is significantly accurate.
- **Availability:** The system is available 24 hours so that the general public can access and use the system.
- **Performance:** The performance of the application is good and the user interface is user-friendly. Anyone who can understand English and have knowledge of web application can use the system.
- **User Satisfaction:** The system is designed to satisfy the user’s needs and requirements.
- **Layout:** The system deals with viewing the different layouts of the application like (UI - User Interface & UX – user Experience design).

### 3.1.2. Feasibility Analysis

Feasibility analysis is used to assess the strengths and weaknesses of a proposed project and present directions of activities that will improve a project and achieve desired results. The nature and components of feasibility studies depend primarily on the areas in which analyzed projects are implemented. The feasibility analysis of this project had been carried out. They are as follows:

**i. Technical Feasibility**

The system is technically feasible to implement. The technology used guarantees reliability, accuracy, and gesture recognition. The technical aspects of the system can easily cover cross platforms like mobile, desktop, web, etc. and is easily scalable. The technical manpower is also easily available.

**ii. Operational Feasibility**

Operational feasibility refers to whether a system will be used effectively after it has been developed. Operational feasibility is the ability to utilize, support, and perform the necessary tasks of a system or program. It includes everyone who creates, operates, or uses the system. To be operationally feasible, the system must fulfill a need required by 6 the business. An example of an operational feasibility study, or the fourth type, analyzes the

inside operations of how a deemed process will work, be implemented, and how to deal with change resistance and acceptance.

### iii. Economic Feasibility

Economic Feasibility is used to determine the financial resources of the project. It measures all costs incurred in the development of the new system. The development of our system is within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased. It focuses specifically on the financial aspects and possible questions raised are:

· System cost-effective

· Outweigh cost

· Estimated cost.

### iv. Schedule Feasibility

The project schedule is feasible as the development is organized into smaller modules and implemented using an iterative approach, allowing continuous progress and timely completion. Core functionalities are prioritized first, followed by incremental addition of features, testing, and refinement. With proper task distribution and regular development cycles, the project can be completed within the allocated academic timeframe without significant delays. The project timeline for this project is given by following Gantt chart.

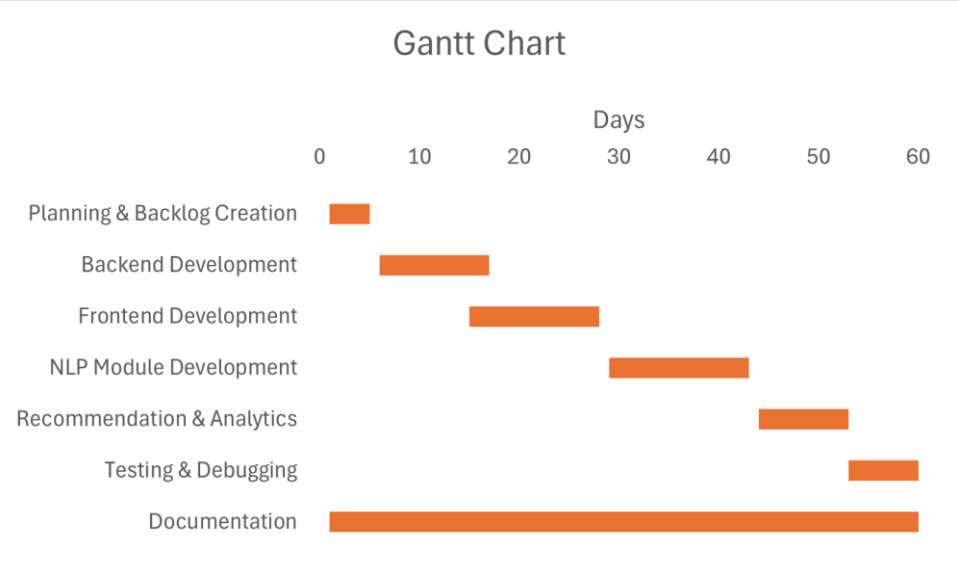



**Figure 2: Gantt Chart**

### 3.1.3. Data Modeling

Data modeling is an essential step in designing a Sign Language Recognition system. It involves defining the structure and organization of data that the system will use, manage, and process. In this project, data modeling is critical for handling video frames, hand landmarks, annotations, training data, and sign language dictionaries. Here's a high-level data modeling overview for the project through ER Diagram:

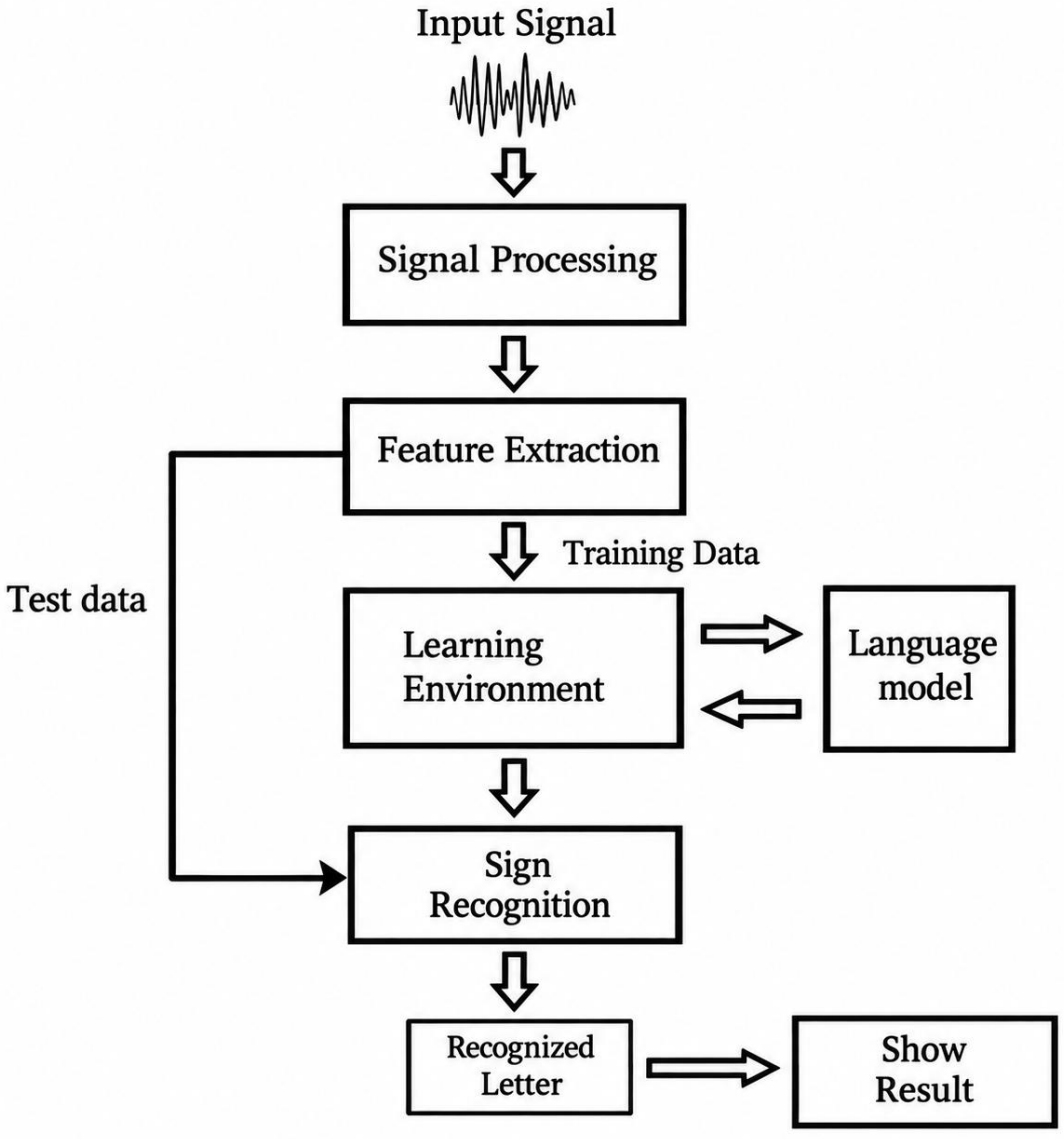



**Figure 3: Data Modeling of Sign Language Translator**

### 3.1.4. Process Modeling

Modeling of structured activities or tasks that produce a specific product for particular users or customers is process modeling. This system involves different processes like the login process, prescription uploading process, and process.

## 3.2. System Design

During the development, we used the following diagrams to understand the requirements of the system.

### 3.2.1. Architectural Design

System design is the process of defining the elements of a system such as the architecture, modules, and components, the different interfaces of those components, and the data that goes through the system. System design is meant to satisfy the specific needs and requirements of a business or organization through the engineering of a coherent and well-running system.

### 3.2.2. Interface Design

The high-level design of a Sign language Recognition system using Python contains three basic components:

**Input Segment:** In a separate area of the interface, in this area user make the sign of different character and they are saved on separate storage which will be useful for training.

**Video Feed:** A real-time video feed from the user's camera displays on the main screen. This feed allows users to see themselves signing in sign language.

**Sign Language Recognition Output:** A dedicated section of the screen displays recognized signs or phrases in real time. This is where users can see the system's interpretation of their sign language.

### 3.2.3. Process Design

Process design illustrates how each process in the system communicates with each other to perform designated tasks. Different figures are used to illustrate different processes of communication:

a) **Flowchart**
   Flowcharts are graphical representations of workflows of stepwise activities and actions with support for choice and iteration. It shows the overall flow of control as a form of picture.

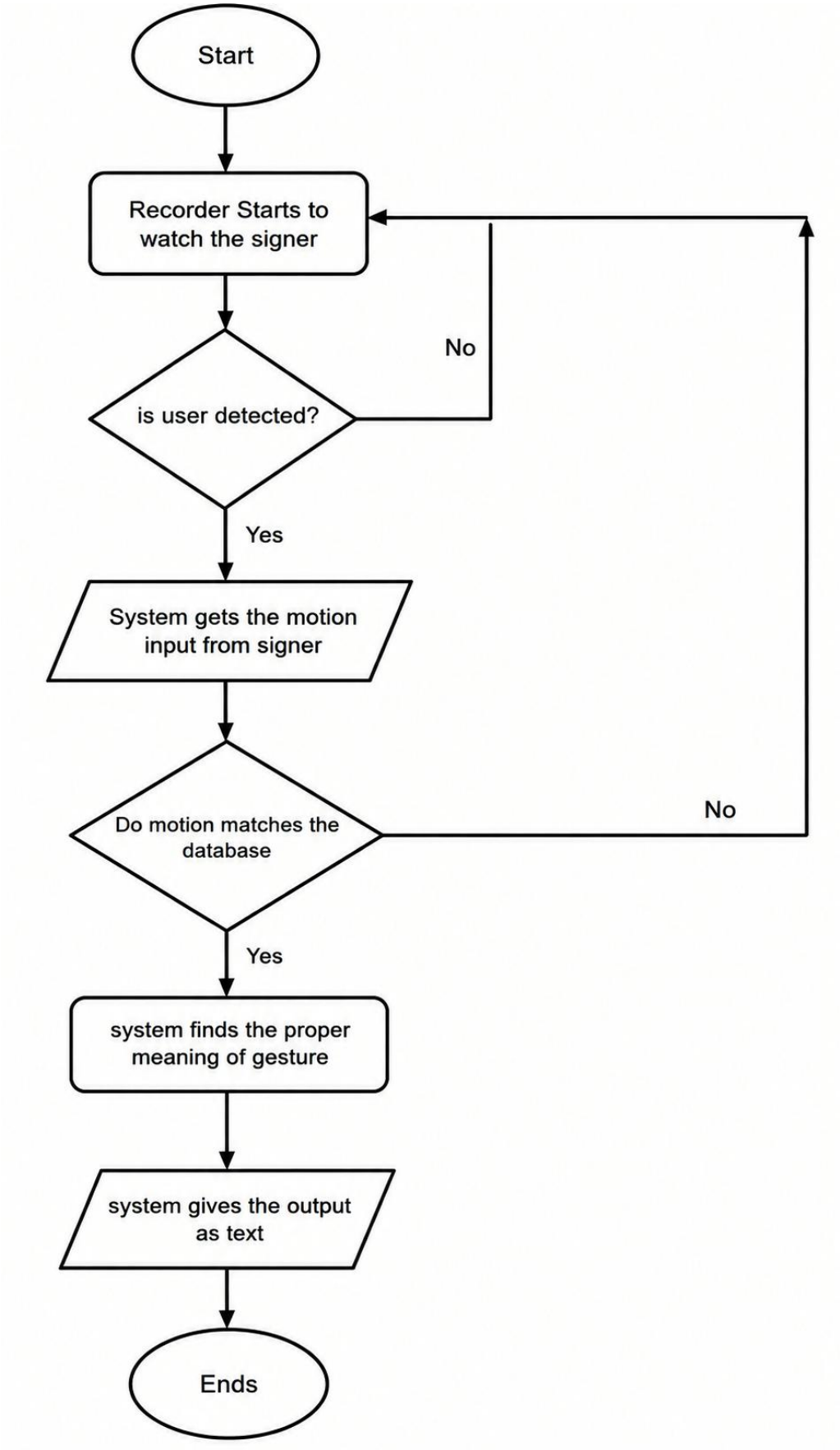



**Figure 5: Flowchart of Sign Language Translator**

# CHAPTER 4: IMPLEMENTATION AND TESTING

## 4.1. Implementation

For the development of the project, the waterfall model suits perfectly. The waterfall model is a sequential design process, used in software development processes, in which progress is seen as flowing steadily downwards through the phases of planning, analysis, design, implementation, and testing.

The planning phase is used for understanding why a system should be built and determining how the project team will go about building it. The analysis phase answers the questions of who will use the system, what the system will do, and where and when it will be used. During this analysis phase, the project team investigates any current system, identifies improvement opportunities, and develops a concept for the new system. The design phase decides how the system will operate in terms of the hardware, software, and network infrastructure that will be in place. In the implementation phase, a system is actually created and this phase usually gets the most attention, because for most systems it is the longest and most expensive single part of the development process. The final phase is the testing phase, where the system after implementation needs to be tested to make it error-free.

### 4.1.1. Tools Used

**Python**: Python is a versatile programming language commonly used for machine learning, computer vision, and general software development. It serves as the primary programming language for the project.

**OpenCV (Open Source Computer Vision Library)**: OpenCV is a popular open-source library for computer vision and image processing. It provides tools and functions for capturing, processing, and analyzing video and image data from cameras.

**MediaPipe**: MediaPipe is a framework developed by Google that offers a range of pre-trained machine learning models for various vision and multimedia tasks, including hand gesture recognition. It simplifies the development of computer vision applications.

**NumPy**: NumPy is a fundamental library for numerical computing in Python. It is used for the efficient handling of multidimensional arrays, which is essential for processing image data and preparing it for machine learning.

**TensorFlow and Keras**: TensorFlow is an open-source machine learning framework developed by Google, and Keras is a high-level deep learning API that runs on top of TensorFlow. These tools are used for building, training, and deploying deep learning models, including LSTM networks.

**Table 1: Tools used for Sign Language Translator**

| **Tool's name** | **Purpose** |
|---|---|
| Visual studio code | IDE for developing software |
| Python, Tensor flow | Server-side programming |
| Python | Client-side programming |
| Microsoft word | Documentation |

### 4.1.2. Implementation Details of Modules

Implementing a Sign Language Recognition project involves several key modules that work together to capture, process, recognize, and translate sign language gestures. Below, I'll provide an overview of the implementation details for each module in the project

**i. Data Collection Module:**

- Capture real-time video feed from a camera (e.g., webcam).
- Continuously capture video frames and convert them to image data for processing.

ii. **Hand Detection Module:**

- Utilize a hand detection model from MediaPipe to identify and extract hand landmarks in each frame.
- Create bounding boxes around detected hands for further analysis.
- Apply filters to remove noise and improve hand landmark accuracy.

iii. **Data Annotation Module:**

- Annotate the detected hand landmarks and gestures. Associate each frame with the corresponding sign language gesture.
- Store annotated data in a structured format for training and evaluation.

iv. **Machine Learning Module:**

- Prepare the annotated data for machine learning by converting it into a suitable format (e.g., NumPy arrays).
- Split the dataset into training and testing sets for model evaluation.
- **:** Design and build an LSTM-based neural network for recognizing sign language gestures.
- Train the neural network using the training dataset and evaluate its performance.

v. **Sign Language Dictionary Module:**.

- Create a dictionary that maps recognized sign language gestures to their corresponding meanings or text translations.

**vi. Real-time Recognition Module:**

- Continuously process video frames, extract hand landmarks, and feed them to the trained recognition model.
- Display recognized signs and their translations in real-time.
- Update the translation output as new signs are recognized, creating a seamless conversation.

## 4.2 Testing

Software Testing is the process of testing the functionality and correctness of software. Software testing is an empirical technical investigation conducted to provide stakeholders with information about the quality of the product to the context in which it is intended to operate. This includes but is not limited to, the process of executing a program or application with the intent of finding errors.

**Unit Testing**

Unit testing refers to the testing of individual units/components of software. The purpose is to validate that each unit of the software performs as designed. A unit is the smallest testable part of any software. It usually has one or a few inputs and usually a single output.

**System Testing**

System testing tests a completely integrated system and the outputs generated by it to verify that the system meets its requirements. It is also used to check logic changes made in it with the intention of finding errors. This process helps in validating the system by testing

the system as a whole that covers each module of the application, database specifications, and underlying configurations. Plagiarism Comparison passes the system testing and is ready for real-world implementation.

## 4.3 Test Cases

**Table 2: Test Case for Sign Language Translator**

| | |
|---|---|
| Use Case ID | 1 |
| Test Case Name | Check valid hand gestures from database |
| Test Case Description | The Gesture of hands should be in a fixed position. |
| Steps | 1. Open the application from the command prompt.<br>2. Provide proper hand gesture |
| Expected Results | If it satisfies the validation rule, it gives the result. |
| Actual Results | As expected, |

# CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATION

## 5.1. Conclusion

This project was undertaken to solve the underlying issue faced by hearing and speech-impaired people. They often don't even stand a chance in the competitive global arena because of communication hurdles.

This project, however, helps eradicate the social stigma of them not being able to participate in many domains and successfully gives them the confidence to stand upright in any field they want.

The application provides the necessary platform to communicate with much ease and gives them the ability to interact without any external help. The need for an interpreter is eradicated, and the smooth flow of a conversation is well-developed.

## 5.2. Future Recommendations

The model and text-to-speech can be embedded into a video calling system. Thereby allowing the user to show the gestures and the receiver on the call will receive the message in the form of text or speech. While the receiver responds, the message will be relayed to the hearing/speech-impaired user via text (subtitles).

The system can further be enhanced in the following ways:

- Integrate a relational database (such as MySQL or SQLite) to support multiple users, maintain a gesture dictionary, and store translation history.
- Incorporate user authentication, data encryption, and secure data transfer (e.g., HTTPS/SSL), particularly for cloud-based or multi-user deployment.
- Extend the dataset to cover a more complete sign language vocabulary, including full ASL/NSL (Nepali Sign Language) signs rather than individual letters alone.
- Enhance model performance under varying lighting conditions and complex backgrounds.
- Extend the system into a mobile application to further improve accessibility.

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# APPENDICES

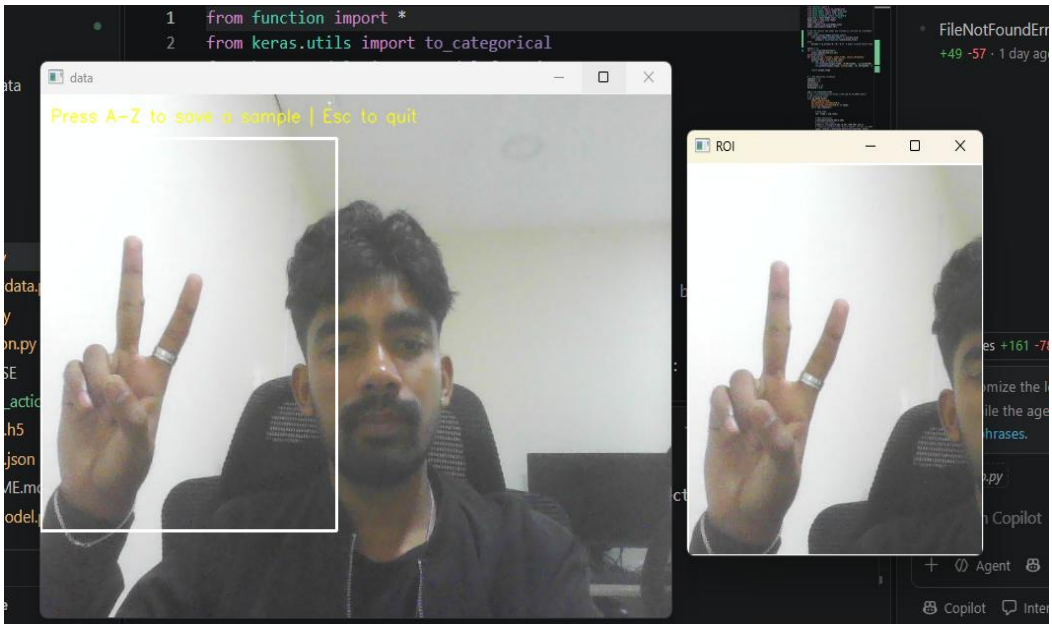



**Figure 6: User Screen**

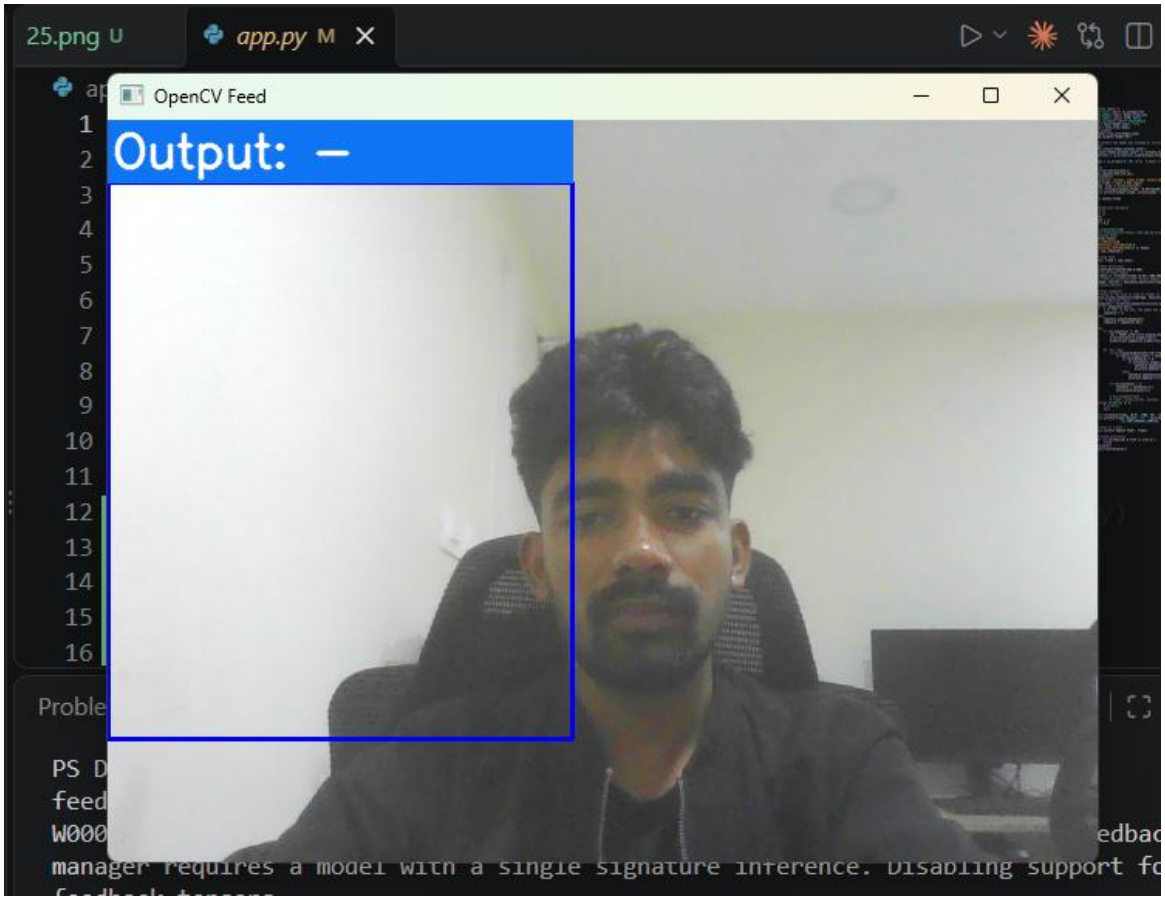



**Figure 7: Running Screen**

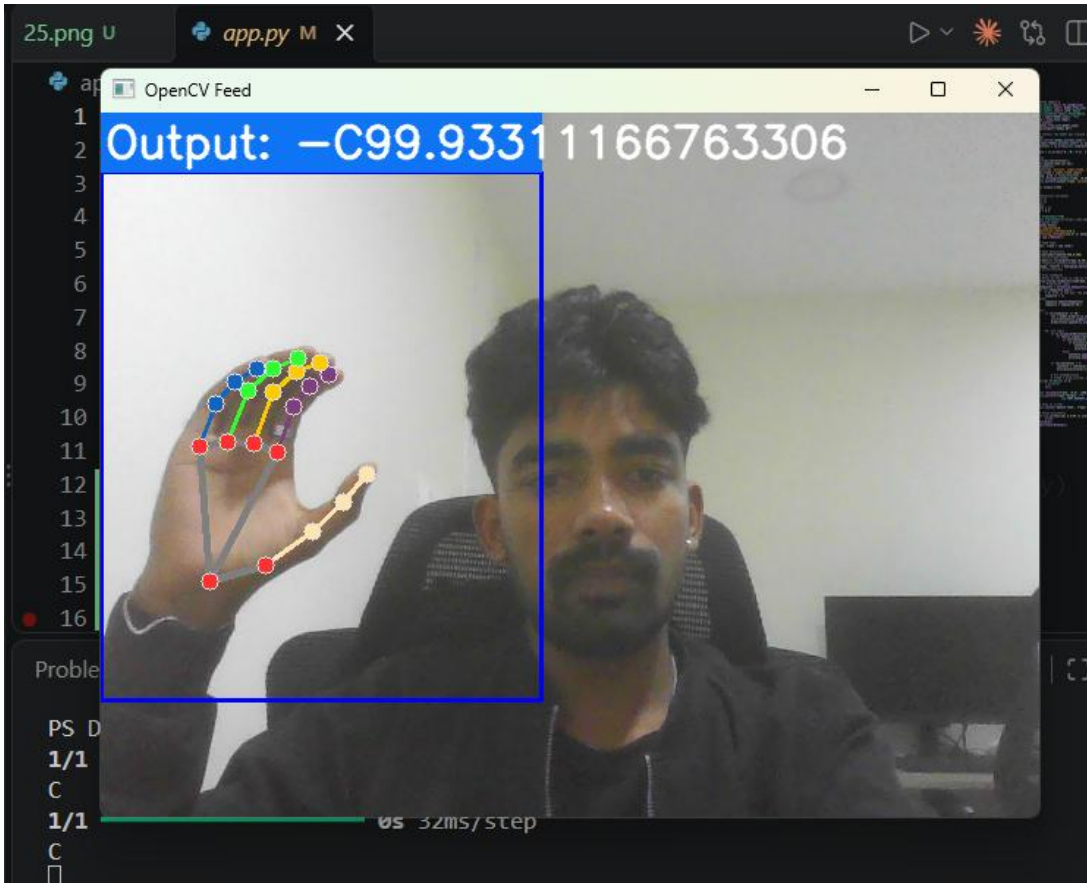



**Figure 8: Output of sign C**

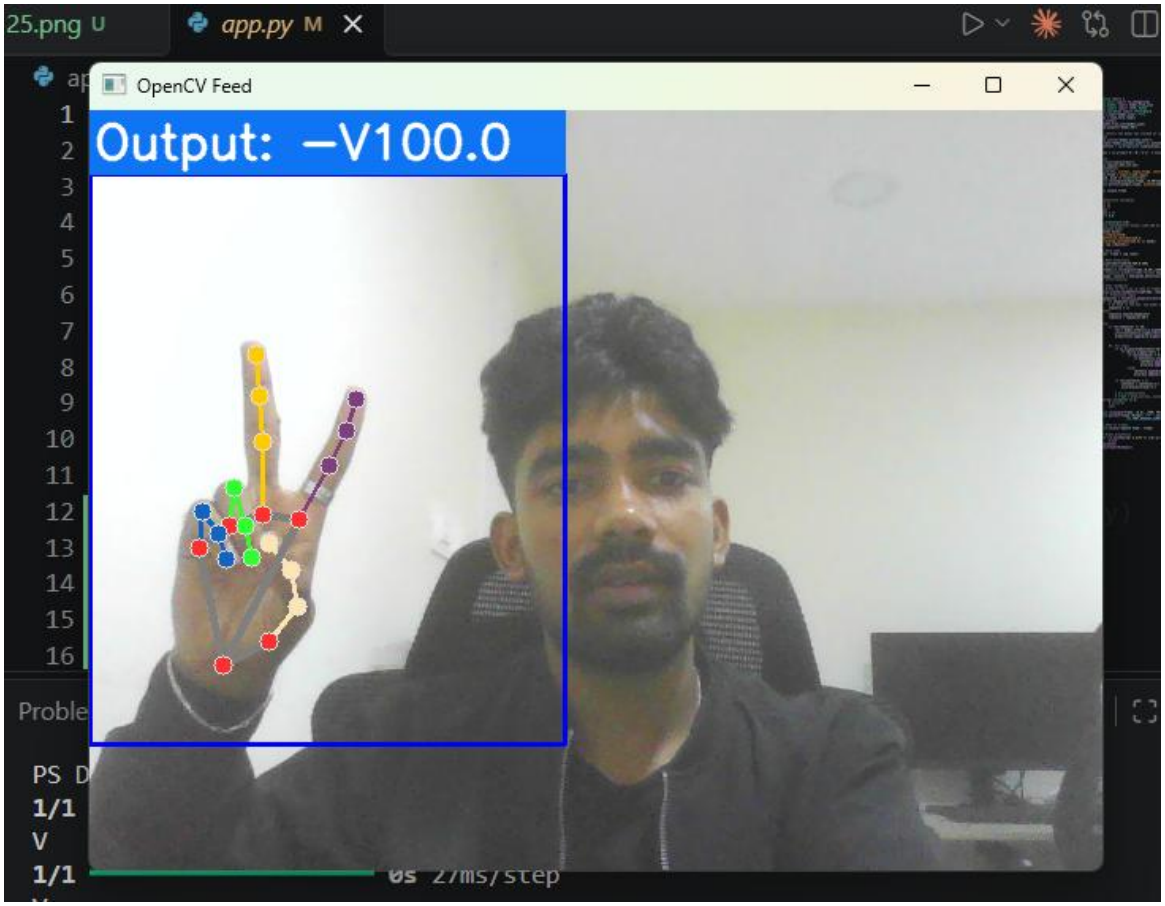



**Figure 9: Output of sign V**

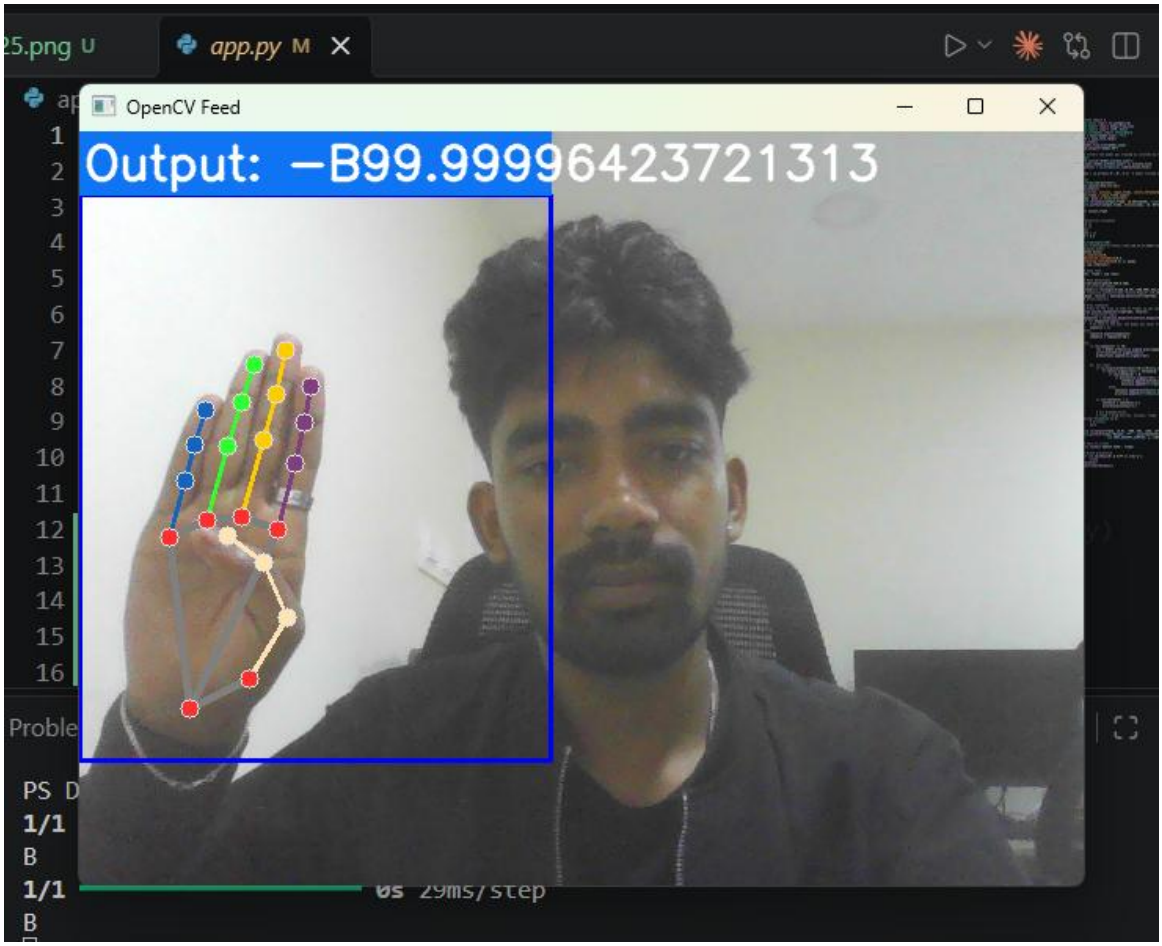



**Figure 10: Output of sign B**

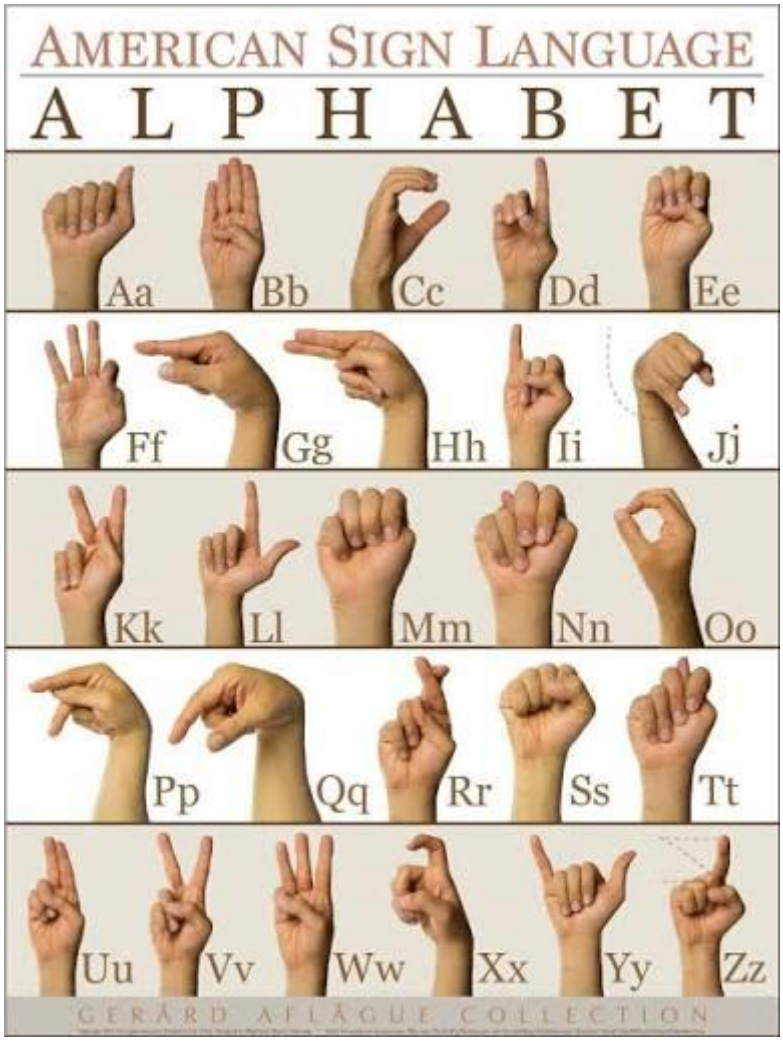



**Figure 11: Gesture Signs**